

Lesson Exemplar in General Science

Quarter 3

LE

12

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Lesson Exemplar for General Science
Quarter 3: Unit 4

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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	2 nd	Quarter	3

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	The learners learn that climate change affects ecosystems and the organisms living in them; and ecosystems are finite and have limited resources.
Performance Standard	By the end of the quarter, learners use examples to describe the relationships among living systems and their components and how the structure of a part relates to its function. They explain the role of cells in perpetuating life and how organ systems work together to maintain optimal health. They conduct investigations on the effects of climate change on local ecosystems.
Learning Competencies	Key Topics Learned in the Previous Grade Level: - Grade 7 (Life Science): Learners studied the components of ecosystems such as producers, consumers, and decomposers. They were introduced to the idea of balance in nature through food chains and food webs, emphasizing how energy flows from one organism to another. - Grade 8 (Life Science): Learners learned about the cycles of matter in ecosystems, including the water, carbon, and nitrogen cycles. They also discussed how human activities such as deforestation and pollution disrupt these cycles and affect ecosystem stability. - Grade 9 (Life Science): Learners examined the interactions among organisms, including competition, predation, and symbiosis, and how these interactions influence population size. They were also introduced to the concept of carrying capacity, which highlighted that ecosystems have limited resources to support populations. - Grade 10 (Earth and Environmental Science): Learners explored environmental issues caused by human activities, such as biodiversity loss, habitat destruction, and climate change. They also learned that population growth directly impacts the availability of resources and contributes to global environmental challenges.



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	These foundational topics from Grades 7–10 serve as scaffolding for this lesson. In this lesson, grade 11 learners <i>analyze information from secondary sources to evaluate the adverse effects of overpopulation locally and globally</i>. Teachers should connect these prior learnings to highlight continuity and avoid redundancy.
II. REFERENCES and MATERIALS (Selecting Resources and Material)	Delizo, M. (2024, July 27). <i>Overpopulation, poor urban development, garbage blamed for NCR floods</i>. ABS-CBN News. https://www.abs-cbn.com/news/2024/7/27/overpopulation-poor-urban-development-garbage-blamed-for-ncr-floods-2152 Department of Education. (2024). <i>MATATAG Curriculum Guide: Senior High School – General Science</i>. Bureau of Curriculum Development. Kurzgesagt – In a Nutshell. (2019, Sept 6). <i>Overpopulation – The human explosion explained</i> [Video]. YouTube. https://www.youtube.com/watch?v=QsBT5EQt348 Our World in Data. (2024). <i>Global Carbon Budget</i>. https://ourworldindata.org United Nations, Department of Economic and Social Affairs, Population Division. (2025). <i>World Population Prospects: Demographic Profiles – Line charts</i>. https://population.un.org/wpp/graphs?loc=900&type=Demographic%20Profiles&category=Line%20Charts United Nations. (2021, July 15). <i>The future of humanity: Population trends decoded</i> [Video]. YouTube. https://www.youtube.com/watch?v=C2U-wMit9WY World Bank. (2025). <i>World Development Indicators</i>. https://data.worldbank.org Materials • Manila paper / chart paper, markers • ICT tools: Kahoot, Mentimeter, Canva, Google Jamboard, projector • Printed worksheets: KWL chart, IPO chart, Fact or Bluff cards, “What If” worksheet • Case posters (India, China, Africa, Amazon, Philippines)



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<i>(These shall be accomplished per topic)</i>	
III. CONTENT <i>(Sequencing Content)</i>	Philippine Ecosystem • Concept of Overpopulation • Local & Global Effects of Overpopulation • Analyzing Secondary Sources of Information • Evaluating and Proposing Solutions
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	At the end of the lesson, the learners will be able to: 1. Define overpopulation and relate it to the concept of carrying capacity in ecosystems. 2. Interpret secondary sources of data on adverse effects of overpopulation. 3. Propose strategies to prevent and mitigate the negative effects of overpopulation in their local area and globally.
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i> This section focuses on selecting learner-centered, evidence-based instructional approaches such as problem-based learning, collaborative tasks, interdisciplinary integration, and technology-enhanced instruction. These strategies are intended to foster active engagement, critical thinking, and adaptability across diverse learning pathways. The chosen approaches and methodologies will be reflected through varied and relevant activities and assessments that emphasize real-world relevance and application, thereby enhancing learner engagement and comprehension.	ANNOTATION <i>*Instruction to the teacher on how to facilitate the activities.</i> <i>*In the Annotation, explicitly explain how the IDF is applied in each part of the lesson</i>



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(Each part shall have 2-3 varied activities)		
A. Activating Prior Knowledge	1. Activating Prior Knowledge Activity 1: Fact or Bluff Objective: To activate learners' prior knowledge about population and ecosystems while correcting misconceptions. Instruction to Teacher: Prepare cards or allow students to raise hands. Read each statement aloud and let students respond Fact/Bluff. Reveal the correct answer and explain briefly. Statements: 1. The Earth has unlimited resources to support human populations. → Bluff 2. Overpopulation can lead to food and water scarcity. → Fact 3. Only animals compete for resources; humans are exempt. → Bluff 4. High population density in cities can cause traffic congestion and pollution. → Fact 5. Overpopulation is only a problem in poor countries. → Bluff 6. Deforestation can be linked to overpopulation. → Fact 7. The Philippines is among the most densely populated countries in the world. → Fact 8. When population grows faster than resources, carrying capacity is exceeded. → Fact 9. Overpopulation has no effect on climate change. → Bluff	<i>This activity encourages active participation while surfacing learners' existing beliefs. By confirming correct answers and clarifying misconceptions, the teacher ensures that students are ready for deeper exploration. This applies IDF's "Activation" strategy, which primes learners' prior knowledge and builds curiosity to connect old concepts (ecosystem balance, resource limits) with the new focus (overpopulation).</i>



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10. Both local and global ecosystems are affected by population growth. → Fact

Activity 2: KWL Chart

Objective: To help learners connect prior knowledge about ecosystems and population with new learning on overpopulation and track their learning progress throughout the lesson.

Instruction to Teacher:

1. Distribute a KWL Chart template (columns: What I Know, What I Want to Know, What I Learned).
2. Ask students to spend 3–5 minutes individually filling in the What I Know column about ecosystems, population, and resource use. Encourage them to recall topics from Grades 7–10 (e.g., carrying capacity, ecosystem balance).
3. Instruct them to complete the What I Want to Know column, listing questions they are curious about regarding overpopulation.
4. Pair students to share their answers, then consolidate in small groups.
5. On manila paper or a digital board (Jamboard/Padlet), ask groups to post 1–2 entries for each column.
6. At the end of the lesson, return to the KWL Chart and ask students to fill in the What I Learned column, connecting new insights with their initial questions.

This activity promotes metacognition by encouraging students to reflect on what they already know and what they want to learn. It also fosters collaboration as they share in pairs and groups. By revisiting the chart at the end, students engage in reflection and self-assessment of their learning.



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	<table> <tr> <th>K (What I Know)</th><th>W (What I Want to Know)</th><th>L (What I Learned)</th></tr> <tr> <td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td></tr> </table>	K (What I Know)	W (What I Want to Know)	L (What I Learned)							
K (What I Know)	W (What I Want to Know)	L (What I Learned)									
	Activity 3: Quick Poll Objective: To gather student perceptions about overpopulation. Instruction to Teacher: Use Kahoot or Mentimeter to ask: 'What is the biggest effect of overpopulation?' Show results as a word cloud. Facilitate a short discussion: highlight the most chosen answers, ask why students picked them, and link their responses to the upcoming lesson. 2. Establishing the Purpose of the Lesson Activity 1: Motivational Questions with Visual Prompts Objective: To spark curiosity by linking overpopulation to learners' everyday experiences. Instruction to Teacher: Project or show images (e.g., traffic in EDSA, crowded markets, polluted rivers, deforested areas).	<i>This activity goes beyond engagement by serving as a diagnostic tool. It gives the teacher a quick snapshot of learners' perceptions and possible misconceptions about overpopulation. It also allows learners to see how their peers think, which stimulates curiosity and discussion.</i> <i>This activity uses visual stimuli to make abstract concepts concrete and relatable. It applies IDF Contextualization by grounding the lesson in local realities and Responsive Instruction by inviting learners to share personal connections.</i>									



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EDSA Traffic



Divisoria Market



Manila Bay



Part of El Nido, Palawan

Guide Questions:

1. What do these pictures have in common?
2. How might population growth contribute to these issues?
3. Which of these problems have you personally experienced in your community?

Activity 2: Video Clip with Guided Reflection

Title: The Future of Humanity: Population Trends Decoded

Link: <https://www.youtube.com/watch?v=C2U-wMit9WY>

The teacher must ask volunteer to share their responses. This activity integrates ICT and multimedia learning, helping learners visualize global data. It applies IDF's Integrative Learning



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	Author / Channel: United Nations Duration: 5:32 minutes Video Description: This video explores global population trends and projections, examining how human population growth has accelerated, what factors influence future trajectories, and potential challenges that come with increasing numbers of people on Earth. It presents key data (birth rates, death rates, population peaks) and discusses environmental, economic, and societal implications. Guide Questions: 1. What are the main trends in human population growth shown in the video (past, present, projected)? 2. What factors does the video identify as influencing population growth? 3. According to the video, when might the global population peak and why? 4. What environmental or societal challenges were mentioned that come with increased population? 5. In what ways could these challenges be similar to or different from those affecting the Philippines? 6. Based on the video, which strategies or policies could help manage population growth sustainably? Activity 3: Think-Pair-Share (Local Ecosystem Focus)	<i>by linking local and global contexts, and Reflective Practice as students process the significance of what they saw.</i> <i>This activity allows learners to construct meaning collaboratively while building confidence before speaking in front of the class. It applies IDF's Social Constructivism by leveraging peer discussion, and Activation by</i>
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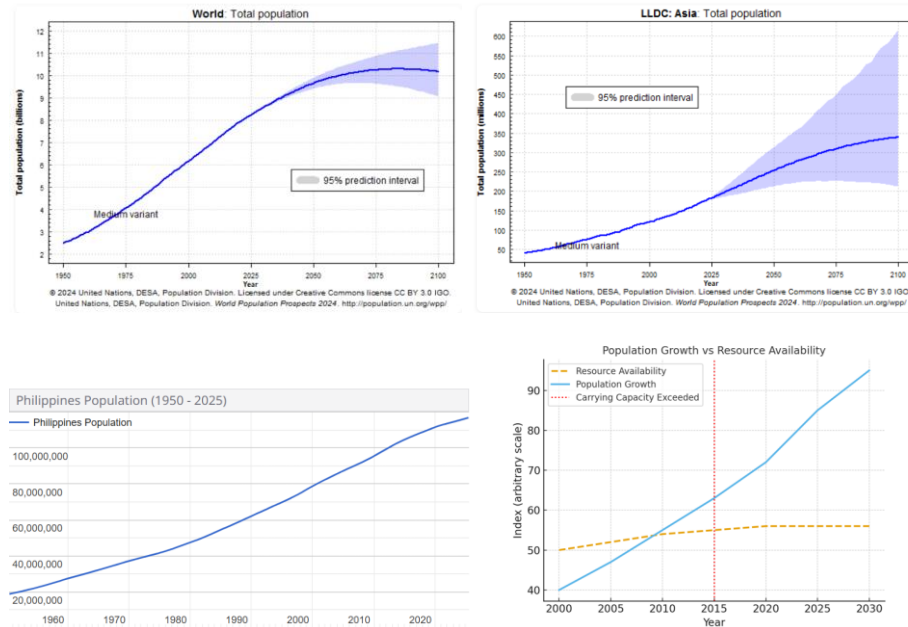
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	Objective: To connect prior thinking to the lesson purpose through collaboration. 1. Pose the prompt: “How do you think population growth affects ecosystems in your community?” 2. Let the students think for 2 minutes to reflect individually then share ideas with seatmates noting at least one common idea and finally share their consolidated idea to the class.	<i>helping learners verbalize their existing perceptions as a bridge to new concepts.</i>
B. Instituting New Knowledge	1. Presenting Examples Activity 1: Mini-Lecture with Interactive Graph Analysis Objective: To introduce population growth trends and connect them to carrying capacity and resource availability. Instruction to Teacher: 1. Present line graphs comparing Philippine population growth and world population growth. 2. Distribute a partially filled for students to complete during the lecture. 3. Pause at key points and ask comprehension question 4. Ask 1–2 students to explain the completed diagram to the class.	<i>This blends direct instruction (teacher’s role) with active student participation (diagram completion). It applies IDF’s Constructive Alignment by ensuring examples directly match the lesson objective of analyzing overpopulation effects. It also uses Scaffolding, giving learners a structure to interpret new information without overwhelming them.</i>



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Sample Diagrams:



Guide Questions:

1. What trend do you observe in the Philippines' population growth compared to global trends?
2. What does it mean when the population growth line goes above the resource availability line?
3. How is the concept of carrying capacity illustrated in the graph?



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	4. What possible consequences occur if population exceeds carrying capacity? 5. Based on the graph, why is it important to manage population growth? Activity 2: Reading Excerpt with Guided Questions (Local Context) Objective: To analyze real-world examples of overpopulation from secondary sources. Instruction to Teacher: 1. Distribute printed copies of the article “<i>Overpopulation, poor urban development, garbage blamed for NCR floods</i>” to each student or group. Title of article: Overpopulation, poor urban development, garbage blamed for NCR floods Author / Publisher: Michael Delizo, ABS-CBN News Published: July 27, 2024 Link: https://www.abs-cbn.com/news/2024/7/27/overpopulation-poor-urban-development-garbage-blamed-for-ncr-floods-2152 Description: This article discusses how Metro Manila’s overpopulation, poor urban planning, and mismanagement (e.g. waste disposal) exacerbate flooding in the National Capital Region. It cites how drainage systems get	<i>This activity immerses learners in authentic, context-rich secondary source reading, helping them see the real impact of overpopulation locally—fulfilling the Contextualized Inquiry principle of IDF.</i> <i>By annotating and discussing, students practice active reading and critical thinking, not just passive consumption.</i> <i>The teacher’s role in guiding and projecting excerpts fosters responsive instruction, as you can scaffold connections between the text and the lesson objectives.</i> <i>The group presentation and class discussion builds social constructivism (students learn from one another’s insights), and helps link new knowledge (local effects) to the broader objective of analyzing overpopulation impacts.</i>
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- overwhelmed, informal settlements block waterways, and infrastructure cannot keep up with population pressures.
2. Give them 5–7 minutes to read silently (or you may read a selected excerpt aloud).
 3. Provide the guided questions below. Ask students to annotate the text (underline evidence, write short notes) as they answer.
 4. In groups of 3–4, learners compare their answers, discuss differences, and highlight the strongest evidence.
 5. Each group presents 1–2 key points (effect + cause chain) to the class.

Guided Questions:

1. What specific problems in Metro Manila are attributed in the article to overpopulation?
2. What role does infrastructure (e.g. drainage, waste disposal) play in making those problems worse?
3. Which populations or areas are most vulnerable according to the article?
4. How do the problems described connect to ecosystem services (e.g. flooding affecting soil, water, habitat)?
5. What solutions or management suggestions does the article mention or imply?

This activity encourages active movement, peer collaboration, and comparative analysis. It



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	6. From your knowledge and from this article, what are the trade-offs or challenges in addressing these problems? Activity 3: Gallery Walk (Global Perspectives) Objective: To expose learners to diverse examples of overpopulation effects across different regions. Instruction to Teacher: 1. Prepare 4 posters with short case studies (e.g., India's urban congestion, China's resource strain, Africa's food security, Amazon deforestation). 2. Divide class into groups and assign each group a starting station. 3. Groups rotate every 5 minutes, reading the case and writing one <i>effect of overpopulation</i> on sticky notes, which they attach to the poster. 4. After all rotations, groups return to seats and the teacher leads a debrief, asking: • Which effects are common across countries? • Which are unique to certain regions? • How do these global cases relate to the Philippine context? Suggested Posters:	<i>applies IDF's Multiple Learning Styles Engagement (visual, kinesthetic, social), while reinforcing integrative learning by comparing local vs. global contexts. It also promotes student agency as they co-construct understanding.</i>
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A. India-Urban Congestion



B. Philippines -Flooding



C. Amazon- Deforestation



D. Africa – Food Insufficiency



E. China – Resource Strain

2. Discussing New Concept

Activity 1: IPO (Input-Process-Output) Chart Analysis

Objective: To analyze the causes, processes, and effects of overpopulation using multimedia and secondary sources.

Instruction to Teacher:

Let the students watch the Video after which they will answer the IPO chart with 3 columns:

- **Input (Causes)** – e.g., high birth rates, migration, lack of planning
- **Process (How it happens)** – e.g., overcrowding, pollution, slum development

After watching the video or reading reports, ask students to complete Input-Process-Output chart (Causes → Processes → Effects).

This activity engages analysis and integration by having learners break down complex processes into inputs, mechanisms, and outputs, which supports scaffolding and structured thinking. It is aligned with the objective of analyzing information from secondary sources. Providing a template helps learners organize their thinking (scaffolding). Having them present and compare charts fosters social constructivism—they learn from peers' perspectives. The teacher's role in facilitating discussion ensures responsive instruction and helps connect the general video content to the local Philippine context, reinforcing contextualized learning.



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	◦ Output (Effects) – e.g., food insecurity, habitat destruction, flooding Title: Overpopulation – The Human Explosion Explained Link: https://www.youtube.com/watch?v=QsBT5EQt348 Channel / Author: Kurzgesagt – In a Nutshell Duration: 6:39 minutes Description: This animated video explores how the human population has grown dramatically, what factors contribute to it, and what possible consequences could follow. It presents key data trends and invites reflection on sustainability. Activity 2: Data Interpretation Challenge Objective: To develop critical thinking through analysis of statistical data related to population growth. Instruction to Teacher: 1. Provide a graph or table showing Philippine population density vs. agricultural land or global population vs. carbon emissions (sources: World Bank, UN, Our World in Data). 2. In pairs, students answer guided questions: ◦ <i>What trend do you notice in the data?</i> ◦ <i>What relationship exists between population and resources?</i>	<i>This activity develops students' data literacy and critical reasoning, aligning with the objective to "analyze information from secondary sources." The teacher scaffolds thinking by providing structured questions and visual prompts. Through group discussion and presentation, learners engage in social constructivism, comparing interpretations and refining their understanding. Live annotation during sharing helps responsive instruction, as the teacher can highlight misinterpretations or guide deeper insights. Because the data is real and relevant to overpopulation issues, this activity supports authentic, contextualized learning.</i>
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○ *What might happen if the trend continues?*

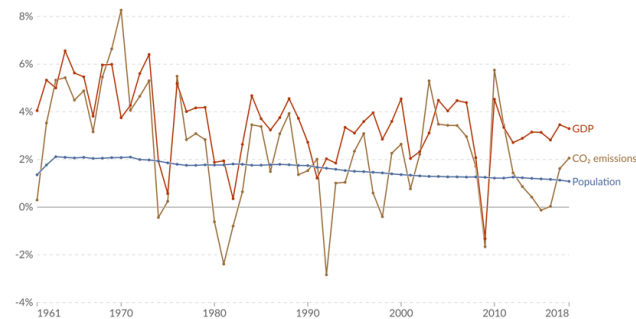
3. Ask groups to make 1 conclusion and 1 recommendation based on the data.

4. Present group answers to the class.

Sample graphs and tables:

Annual change in GDP, population and CO₂ emissions, World

Percentage change in gross domestic product (GDP), population, and carbon dioxide (CO₂) emissions.



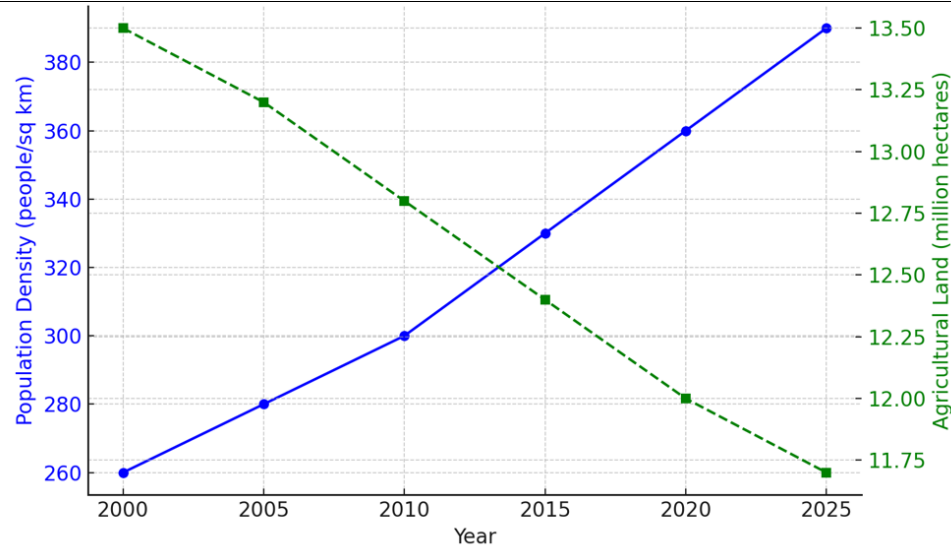
Data source: United Nations Population Division, Eurostat, national statistical offices, and United Nations Statistics Division, via World Bank (2025); National statistical organizations and central banks, OECD national accounts, and World Bank staff estimates (2025); Global Carbon Budget (2024)

Note: GDP figures are adjusted for inflation.

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY



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Philippines: Population Density Vs. Agricultural Land

3. Developing Mastery

Activity 1: Matching Game – Effects and Consequences

Objective: Reinforce understanding by linking overpopulation effects with their consequences on ecosystems and society.

Instruction to Teacher:

This activity uses reinforcement and practice, essential in IDF's scaffolding for mastery. By actively matching cause-and-effect pairs,



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1. Prepare cards (or digital slides if ICT-based). Each card either lists an *effect* (e.g., “Rapid urbanization”) or a *consequence* (e.g., “Formation of slums”).
2. Distribute shuffled cards to groups.
3. Groups must match each effect card with its correct consequence.
4. Afterward, groups explain one matched pair to the class.

students develop stronger connections, moving from recognition to explanation.

Causes (10 cards)

1. High birth rates and limited family planning programs
2. Rural-to-urban migration
3. Poor waste management in densely populated areas
4. Expanding human settlements into forested areas
5. Increased demand for food supply
6. Heavy traffic congestion in metropolitan areas
7. Lack of proper sanitation infrastructure
8. Overcrowded housing conditions
9. Dependence on fossil fuels and industrial expansion

Effects (12 cards)

1. Deforestation and loss of biodiversity
2. Spread of diseases in slums
3. Air pollution and respiratory illnesses
4. Overfishing and depletion of marine resources
5. Decline in agricultural land per capita
6. Urban flooding due to clogged drainage
7. Reduced access to clean drinking water
8. Longer commuting times and decreased productivity
9. Desertification in arid lands



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	10. Climate change intensifying natural disasters in urban areas	10. Increase in unemployment and poverty 11. Strain on public services (healthcare, education, transportation) 12. Food scarcity and malnutrition	
	Activity 2: Structured Question-and-Answer Objective: Deepen mastery by prompting learners to justify, explain, and evaluate concepts related to overpopulation. Instruction to Teacher: 1. Pose open-ended “Why” and “How” questions to the class. Examples: ○ <i>Why does overpopulation put pressure on food security?</i> ○ <i>How might overpopulation contribute to climate change?</i> ○ <i>Why do effects differ between developed and developing countries?</i> 2. Ask students to respond first in pairs, then invite volunteers to share. 3. Teacher clarifies and expands on student answers to correct misconceptions.		<i>This activity develops critical thinking and ensures learners move beyond surface knowledge. It aligns with IDF’s principle of progressive mastery—from guided practice toward independent reasoning. Pair-discussion before class-sharing also reflects social constructivism, supporting learners to refine ideas collaboratively.</i>



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	Activity 3: Concept Mapping – Overpopulation Web Objective: Demonstrate mastery by organizing knowledge of causes, processes, and effects into a visual framework. Instruction to Teacher: 1. Provide each group with a large sheet of paper (or digital tools like MindMeister, Canva, or Google Jamboard). 2. Ask them to create a concept map starting with “Overpopulation” at the center. 3. Branch into <i>Causes</i> (e.g., high birth rate, poor family planning), <i>Processes</i> (urban congestion, waste accumulation), and <i>Effects</i> (pollution, flooding, deforestation, hunger). 4. Groups present their maps to the class. 5. Teacher synthesizes common links and highlights missing or unique connections.	<i>Concept mapping is a visual scaffolding tool that consolidates student learning. It promotes higher-order organization and synthesis, key aspects of developing mastery in IDF. The collaborative aspect reinforces peer-to-peer learning, while presentations allow teacher feedback to ensure conceptual accuracy.</i>
C. Demonstrating Knowledge and Skills	1. Finding Practical Application Activity 1: Structured Debate – “Is Overpopulation the Root Cause of Environmental Problems?” Objective: To apply critical thinking and communication skills in analyzing whether overpopulation is the primary driver of environmental challenges. Instruction to Teacher:	<i>The debate builds practical communication, persuasion, and critical analysis skills. It aligns with IDF’s application principle, as students must apply theoretical knowledge to real-world arguments. The structured format ensures fairness and scaffolds students’</i>

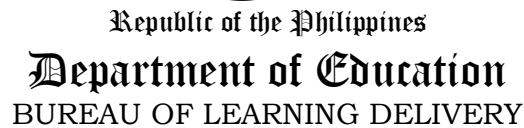


	1. Divide the class into two debating teams (Affirmative and Negative) and a group of moderators/judges. 2. Provide each team 10 minutes to prepare their arguments using notes, case studies, and class learnings. 3. Debate format: - o Opening statements (2 minutes each team) - o Rebuttals (2 minutes each) - o Open floor (moderator poses 2–3 guide questions) - o Closing statements (1 minute each) 4. Moderator Guide Questions: - o <i>What is the strongest evidence linking overpopulation to environmental problems?</i> - o <i>Are there alternative root causes aside from population?</i> - o <i>Which solutions are more realistic—population control or better resource management?</i> 5. Judges (or whole class) vote on which team presented stronger arguments. Activity 2: “What If” Scenario Worksheet Objective: To reflect on real-life implications of population growth in community settings. Instruction to Teacher: 1. Distribute a What If worksheet with scenarios such as:	<i>reasoning, while the guide questions help moderators manage flow—demonstrating social constructivism and contextualization.</i> <i>This activity promotes personal reflection and community connection, aligning with IDF’s contextualized application. It pushes learners to consider local impacts of global issues and propose practical solutions, reinforcing integrative learning.</i>
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	○ <i>What if the population of your town doubled in 10 years?</i> ○ <i>What if agricultural land in your province was reduced by half?</i> ○ <i>What if waste management did not improve despite more people?</i> 2. Ask students to work individually, writing their predicted outcomes and possible solutions. 3. In groups of 3–4, they share their answers and pick the most pressing scenario. 4. Groups present their solutions briefly to the class. 2. Making Generalization Activity 1: 3-2-1 Summary: Write 3 things learned, 2 interesting facts, 1 question they still have. (In this activity, instruct the learners to 3 things they learned about overpopulation and ecosystems, 2 things that surprised them or changed their perspective, 1 action they can do in their community related to population issue.) Activity 2: Round Table Synthesis Objective: To promote collaborative articulation of key concepts. Instruction to Teacher: 1. Divide students into small groups (5–6 members). 2. Provide chart paper and markers.	<i>This activity builds metacognition by making students pause and reflect on their learning journey. It applies IDF reflective practice, helping learners connect knowledge to personal insights and actionable steps.</i> <i>During synthesis, if in case there is/are misconceptions, make sure to correct and immediately give feedback to reinforce learning. This activity uses social constructivism by requiring groups to co-create</i>
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	3. Ask groups to discuss and write a 2–3 sentence generalization that answers: <i>“What is the overall impact of overpopulation on ecosystems locally and globally?”</i> 4. Groups post their papers on the board or wall. 5. Conduct a quick gallery read, then learners synthesizes. Activity 3: Teacher-led Concept Integration Objective: To ensure students’ generalizations align with scientific accuracy. Instruction to Teacher: 1. Review students’ reflections and group generalizations. 2. Create a concept map (on the board or projector) summarizing key points: ○ Causes (inputs) ○ Processes (mechanisms) ○ Effects (outputs) ○ Solutions (applications) 3. Connect their insights to the performance standard (using examples to explain relationships and solutions). 3. Evaluating Learning Activity 1: Fact or Bluff – Misconception Check	<i>a concise generalization. It encourages collaborative meaning-making, a key IDF application for generalization.</i> <i>This ensures constructive alignment by explicitly linking student-generated ideas to the competency and standard. Teacher facilitation plays a crucial role in scaffolding, making sure the “big picture” is scientifically valid and aligned with objectives.</i>
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Directions: Write *Fact* if the statement is correct and *Bluff* if it is incorrect.

1. The Earth's resources can support unlimited human population growth.
2. Overpopulation contributes to urban flooding in Metro Manila.
3. Overpopulation can worsen waste accumulation and pollution in cities.
4. Overpopulation can lead to food scarcity and malnutrition.
5. Overpopulation is a problem only in developing countries.
6. Increasing population density puts pressure on agricultural land.
7. Waste mismanagement in overpopulated areas can clog drainage systems and worsen floods.
8. The Philippines is one of the most densely populated countries in the world.
9. Population growth has no connection to climate change
10. Strain on healthcare, education, and transportation is one effect of overpopulation.

Activity 2: Short Situational Quiz (5 Items)

Instruction to Teacher: Read each statement aloud or project them on screen. Ask students to write "Fact" or "Bluff" in their answer sheet. After answering, discuss each statement briefly, clarifying misconceptions.

IDF Alignment: This activity promotes active recall and misconception correction, key aspects of evaluation in IDF. It aligns with the objective of analyzing effects of overpopulation and ensures students distinguish facts from misconceptions.

Possible Answers:

1. *Bluff*
2. *Fact*
3. *Fact*
4. *Fact*
5. *Bluff*
6. *Fact*
7. *Fact*
8. *Fact*
9. *Bluff*
10. *Fact*



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Worksheet:

Directions: Read each situation carefully. Encircle the letter of the best answer.

1. A town's population doubles in 15 years, but agricultural land remains the same. What is the most likely effect?
 - a. Food scarcity
 - b. Improved sanitation
 - c. Increased biodiversity
 - d. Reduced unemployment
2. An urban community keeps growing, but its drainage system remains the same size. Which outcome is most likely?
 - a. Cleaner rivers
 - b. Improved housing
 - c. Flooding during heavy rains
 - d. Increased agricultural production
3. A country has high birth rates and limited family planning programs. What is a possible long-term effect?
 - a. Overpopulation and strain on services
 - b. Reduction in traffic congestion
 - c. Increase in natural resources
 - d. Decrease in population
4. Which of the following is a direct ecological consequence of expanding human settlements into forest areas?

Instruction to Teacher: Administer as a short quiz at the end of the lesson. Check answers collectively by projecting the correct responses and explaining them briefly.

IDF Alignment: This activity serves as a formal evaluation of higher-order thinking, aligned with the competency of analyzing secondary sources. It ensures constructive alignment by directly testing learners' ability to apply concepts in real-life scenarios.

Possible answers:

1. *a*
2. *c*
3. *a*
4. *a*
5. *a*



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- a. Deforestation and loss of biodiversity
 - b. Cleaner air and water
 - c. Increase in soil fertility
 - d. Preservation of habitats
5. Which solution best addresses the problem of waste accumulation in overpopulated areas?
- a. Solid waste management system
 - b. Family planning programs
 - c. Urban migration policies
 - d. Expanding farmland

4. Additional Activities

Activity 1: Create an Infographic

Task:

Design a one-page infographic showing the causes, effects, and possible solutions to overpopulation in the Philippines. Use either paper and markers or a digital tool (e.g., Canva, PowerPoint).

Criteria	Excellent (5)	Good (3)	Needs Improvement (1)	Score (___/20)
Content Accuracy (5 pts)	Information is accurate and relevant.	Minor errors but generally accurate.	Many errors or irrelevant information.	
Organization & Clarity (5 pts)	Well-organized and easy to understand.	Somewhat organized; a few unclear parts.	Disorganized; hard to follow.	

Instruction to Teacher: Encourage creativity and guide students to focus on clarity and accuracy of information. Display selected outputs on the bulletin board or online class platform.

IDF Alignment: This activity strengthens extension and creativity, allowing students to apply knowledge in a visual, communicative way.



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Creativity & Visuals (5 pts)	Very creative; visually appealing.	Some creativity and neatness shown.	Minimal creativity; plain or messy.	
Proposed Solutions (5 pts)	Clear, realistic, and well-explained.	Present but lacking detail.	Missing or vague solutions.	

Activity 2: Opinion Blog / Short Essay
Task:

Write a short blog post or 3–4 paragraph essay answering:
“How does overpopulation affect your community, and what small actions can young people take to address it?”

Criteria	Excellent (5)	Good (3)	Needs Improvement (1)	Score (___/20)
Content & Relevance (5 pts)	Ideas are accurate, relevant, and insightful.	Ideas are mostly accurate and relevant, but lack depth.	Ideas are unclear, inaccurate, or off-topic.	
Organization & Coherence (5 pts)	Well-structured; ideas flow logically.	Some organization; minor lapses in flow.	Poorly organized; hard to follow.	
Clarity of Expression (5 pts)	Clear, concise, and engaging writing.	Understandable but may have awkward phrasing.	Confusing; difficult to understand.	
Personal Reflection & Solutions (5 pts)	Shows deep reflection; provides realistic solutions.	Some reflection; solutions need more detail.	Minimal reflection; no clear solutions.	

Activity 3: Community Survey (Mini-Research)
Task:

Interview 3–5 people in your neighborhood (parents, neighbors,

Instruction to Teacher: Give students 20 minutes to draft, or assign it as homework. Invite volunteers to read their essays aloud. IDF Alignment: This builds reflective and integrative learning, encouraging students to connect lesson concepts with personal and community contexts.

Instruction to Teacher: Guide students to keep questions simple and respectful. Review and discuss findings in class, highlighting common themes.



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	barangay officials) with the question: “What do you think is the biggest effect of overpopulation in our community?” Document the interview process. Summarize responses in a table and share in class.	<i>IDF Alignment: This promotes authentic learning and inquiry, extending the classroom to the community. It aligns with IDF’s application and reflection since students use real-world data to reinforce concepts.</i>
V. ASSESSMENT (Assessing Learnings)	. • Summative Test: Give students 15-item multiple choice. (Please see annex 1 for the summative test and answer key with rationalization.)	
VI. REFLECTION (Feedback and Continuous Improvement)	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>	



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Annex 1. Suggested Assessment

Direction: Read each question carefully and choose your answer from the given options.

1. Which of the following BEST describes overpopulation?
 - a. A sudden decrease in population due to migration
 - b. The movement of people from rural to urban areas
 - c. The process of maintaining balance between people and resources
 - d. A condition where the number of people exceeds the resources available

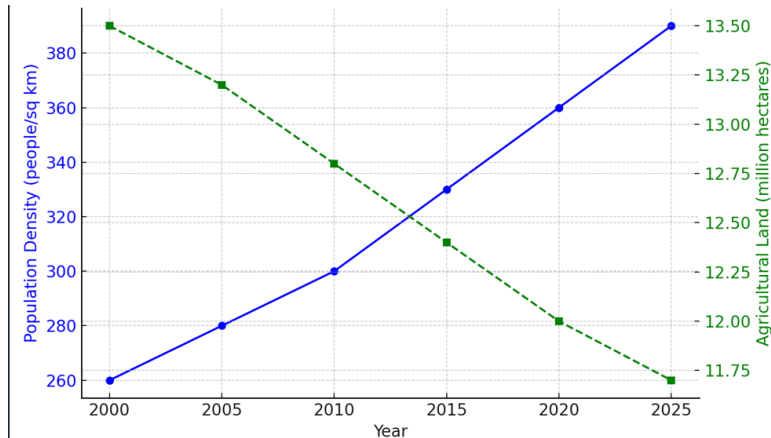
2. Which country is often cited as one of the most densely populated in Southeast Asia?
 - a. Malaysia
 - b. Philippines
 - c. Thailand
 - d. Vietnam

3. Which is NOT a direct effect of overpopulation?
 - a. Increased demand for housing
 - b. Strain on public services
 - c. Enhanced biodiversity
 - d. Food scarcity

4. The graph below shows the Philippines' population density increasing while agricultural land decreases. What conclusion can be drawn?



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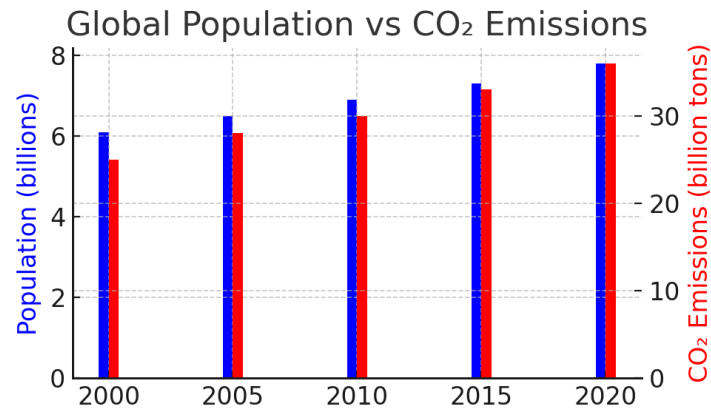


- a. More forests are being preserved
 - b. Overpopulation has no link with agriculture
 - c. Food production will increase due to urbanization
 - d. Agricultural resources may not keep up with population growth
5. A news article reports: "Metro Manila experiences flooding due to clogged waterways from unmanaged garbage in densely populated areas." Which cause-effect relationship is most accurate?
- a. Urbanization → reforestation → less flooding
 - b. Rural migration → water scarcity → improved drainage
 - c. Overpopulation → better sanitation → cleaner waterways
 - d. Overpopulation → waste mismanagement → urban flooding
6. A community has a rapid population increase but its hospitals remain limited. What is the MOST likely consequence?



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- a. Improved health outcomes
 - b. Overcrowding in healthcare facilities
 - c. Decreased population density
 - d. Lower mortality rates
7. Based on secondary data from the World Bank, the Philippines has one of the fastest population growth rates in Asia. What long-term effect could this trend have?
- a. Abundant farmland for all citizens
 - b. Pressure on natural resources and services
 - c. Decreased urban congestion
 - d. Reduced demand for education
8. The graph below shows CO₂ emissions rising as global population increases. What inference is valid?



- a. Emissions decrease with more people
- b. Overpopulation reduces industrial activity
- c. Climate change is unrelated to population growth
- d. Population growth contributes to environmental stress



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9. A survey shows that 60% of respondents believe overpopulation is the root cause of poverty. What does this indicate?
 - a. The public sees a strong link between overpopulation and economic conditions
 - b. There is no connection between population and poverty
 - c. Overpopulation always reduces poverty
 - d. Poverty causes overpopulation
10. Which is the BEST long-term solution to food scarcity caused by overpopulation?
 - a. Affirming to family planning programs
 - b. Allowing agricultural land to become residential land
 - c. Encouraging unlimited urban migration and providing jobs to migrants
 - d. Expanding access to reproductive health and family planning education
11. If the Philippine population continues to rise while water resources remain the same, what is the MOST likely outcome?
 - a. More equitable distribution of water
 - b. Increase in water supply per person
 - c. Water shortages and conflicts over resources
 - d. Improved irrigation systems with policy changes
12. Which of the following reflects an effective community-level adaptation to overpopulation?
 - a. Increasing the birth rate
 - b. Ignoring waste management issues
 - c. Expanding housing without sanitation facilities
 - d. Implementing recycling and composting programs



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13. Evaluate this statement: "Overpopulation is a global problem, not just a local one." Which evidence BEST supports it?
 - a. Only developing countries are affected
 - b. Rural migration solves overpopulation
 - c. Climate change impacts are shared globally
 - d. Population decline in one area increases global resources
14. Which is the BEST example of sustainable solution to overpopulation challenges?
 - a. Strengthening reproductive health programs and sustainable farming
 - b. Allowing uncontrolled industrial expansion
 - c. Large-scale deforestation for housing
 - d. Ignoring climate change policies
15. Imagine your barangay population doubled in 10 years. Which action would BEST help address the impact?
 - a. Enhancing solid waste management and public health systems
 - b. Allowing informal settlements to grow unchecked
 - c. Expanding housing while ignoring sanitation
 - d. Reducing school enrollment opportunities



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Answer Key:

No.	Answer	Rationalization
1	d	Overpopulation occurs when population exceeds available resources.
2	b	The Philippines is among the most densely populated countries in SE Asia.
3	c	Biodiversity is reduced, not enhanced, by overpopulation.
4	d	Increasing population density with shrinking farmland means agriculture may not meet demand.
5	d	More people → more unmanaged waste → clogged waterways → flooding.
6	b	Population growth without more hospitals → overcrowding in healthcare.
7	b	Fast growth strains natural resources and services.
8	d	Population growth increases CO ₂ emissions, linking it to environmental stress.
9	a	Survey shows perception that population growth worsens poverty.
10	d	Family planning is a long-term solution to food scarcity.
11	c	Fixed water resources + more people = shortages and competition.
12	d	Recycling/composting are adaptive, sustainable practices.
13	c	Overpopulation causes global climate impacts, not just local.
14	a	Combines reproductive health and sustainable farming for long-term balance.
15	a	Addressing sanitation and waste management is most effective.